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Context based diversification on keyword search by conceptualization of typed terms of the query

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ABSTRACT

The human era is evolved and dominated through the ultimate intention to know about the Universe and its assets more and more. This in turn persuaded him to gather the immense information of need in the form of theory, tools, intuitions, visuals, and ultimately as the form of abstract objects. In modern Information practice, the art of Conceptualization and the induction of the data driven paradigm identifies the user query intention over a Conceptual network to mold a confident search segment of user choice. The proposed Conceptualization with Typed Terms of Query (CTTQ) approach derives a confident segment of the user query which is contextually measured by means of novel randomized Machine learning algorithms and its surrogates. The Normalized Discounted Cumulative Gain (nDCG) measures of the CTTQ show the usefulness of the query suggestions. It's efficiency improvements are represented with varying impact lexical feature indicator (θ) values.

1. Introduction

Identifying the user intention from the search queries especially for the vague queries at an earlier phase is very challenging in an Information System (IS) based retrieval process. If unable to capture the actual user need of a query and the corresponding data is not properly organized, the search operation over a data set of abundant data is a cumbersome task (Jain et al. (2014), Li, Sun et al. (2015), Hua et al. (2016)). This "Aimer doesn't have a Boomerang" scenario is noted in current marketing trends like Block chain technology, Supply chain management, Information retrieval and Social network analysis. The Artificial Intelligence (AI) transformations interlocked with Machine Learning (ML) technology are used to pinpoint and keep track the targeted objects in the above marketing scenarios as pointed as, the first page ranking in Search Engine Results Page (SERP) is necessary for it to be visible to the target potential customer (Deng et al. (2015), Arpan Kumar et al. (2018), Kushwaha and Kar (2020)). The approach saves the browsing time and enhance the product analysis in digital marketing methodologies like Search Engine Marketing (SEM), Search Engine Optimization (SEO) processes and other allied web services. The ultimate intention of any production team is to cope with the user's desires without depreciating the company profit. In order to bridge the gap between the engineering practice and the production research, a comprehensive study is needed, because the user taste may be sometimes absolutely different from the researcher's concerns Zarindast et al. (2021) as the paper mentioned, the user delight such as lamp color some how always crosses with the researchers choice of ultimate efficiency. The mixed research methodology of the conceptual analytics finds a bias over the situation. It says, there are numerous instances which can impose sudden data diversion to a desired search context. The modern context based search diversification approaches avoid serendipity and espionage by emphasizing on theory building around the objects of needed artifact. The art of theory building has an exclusive role in representing the hypothetical core data in a Big data ecosystem. The data driven paradigm targets on the conceptual drifts happening on the user data, and develops AI-ML surrogate solutions for enhanced data retrieval process. Since the data driven paradigm satisfies the specific industry needs it is more focused on the core product development process and other allied business related issues. The Deep learning approaches utilize the essence of both AI and ML, and extracts significant and relevant information for a sustainable smart industry management. The works of Gandhi et al. (2023) proposed a data driven block chain approach to meet the challenges and opportunities posed by the industry standard 4.0 revolution. Here the digital transformation is spurred

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primarily by the growth of Big data and the use of AI driven decision making paradigm (Dwivedi et al. (2023), Jamwal et al. (2022)).

In almost all instances of Big data and Data mining applications including Information Retrieval (IR), the modern data-driven paradigm focuses on the data objects and the operations on it specified by a business strategy (Jones et al. (2023), Kar et al. (2023), Edstrom et al. (2017), Yu et al. (2015), Liu et al. (2018)). It seeks for a concept drift to divert and guide the search in the right way by using the data-intensive digital environments to communicate, collaborate, and share opinions and decision of vendors and users (Adikari et al. (2021), Ruiz et al. (2014)). The Concept drifts or the context diversion method are the changes to the distribution of the data guided by the fine grained patterns, and it decides the dynamicity of the Big data streams (Nallaperuma et al. (2019), Chen et al. (2011)).

What is the role of theory building around a business artifact? The theory building identifies the current business needs and the solution core data objects, accordance with the industry benchmarks (Kar and Dwivedi (2020), Miranda et al. (2022)). It uses the supporting data and the surrounding theory to trigger the proper methodology and to invoke the industry iteratively. The data driven methodology triggers the Big data fine grained patterns to be evolved through, curating, assembling and aligning lexicon during the sequence and synaptic analysis. The approaches effectively solves the "Elephant and blind men" problem by specifically focusing to the information workload, and theorizing the hidden and observed data objects and their associations according to the industry needs, and threats. The Machine Learning approaches concentrate to set the values of parameters of the algorithms responsible to solve the solutions of a specific problem. Even though the ML covers the comprehensive solutions to a range of problems, the frequent updates, changing business strategies, technological impacts and the conceptual drifts occurred on user needs and resources, compelled it to program on some adequate services which are in the scope of a particular business environment. The data conceptualization is such an ML solution by using the data and its collocation, and it is represented as the measure of semantic relatedness and similarity of terms in the search query. Literally says, Conceptualization is the glue which holds the semantic mental world together. During conceptualization, the semantic types of the query which are evolved from surroundings and personal knowledge can be encoded. The query is measured and represented on account of attributes, concepts, and instances. Concepts and instances contribute more to determine the semantics of a query than attributes (Dou et al. (2015), Ferragina and Scaiella (2010)).

The object based fuzzy ontology construction of IR considers both global and local contexts of the query on a conceptual network. The works of (Hua et al. (2015), Jain et al. (2021)) mention about the conceptual query expansion. The approach derives an efficient membership degree to the data units, in which the data units are in the form of documents belong to a large data corpus or merely a query peace collocated on the keywords or tweets (Zhao and Mao (2017), Sijin and Champa (2020)). The word embedding methods effectively calculate the membership degree between the key words or base term cluster and the data corpus (Demir and Ozgur (2014), Huang et al. (2017)). This way of guided query reformulation is different from the keyword cover of Deng et al. (2015), where the domain specific popular web rates are the indicator of a decision.

The Conceptualization with Typed Terms of the Query (CTTQ) model identifies the concept vector of the corresponding keywords, identifies the shared segments and cycles to make the CN more object specifically to drive the given search query. It uses the Maximal Clique Cycle Count (MCCC) algorithm to identify the maximal clique for the query segments and counts the number of cycles formed among the concepts and their associated vectors of the given Concept Network (CN). The precision of text segmentation is improved with MCCC approach and Instance ambiguity is avoided by a threshold based pairwise model with a weighted voting process. The model identifies the Smallest Lowest Common Ancestor (SLCA) nodes of the original query keywords from the XML dataset by referencing the intention cluster or a query's Fuzzy Set (FS). The SLCA nodes act as anchor nodes to hold the query intention since it holds the contextual terms and these typed contextual terms can easily explore the conceptual meaning of the query. The XRANK approach or XQuery language can be used to locate the anchor nodes Guo et al. (2003). The Anchor based Evaluation of Semantic Pattern (AESP) algorithm can distribute the query and generated query's SLCA nodes to the dedicated distributed processing nodes such as Hadoop Data nodes by referring the intention cluster.

Table 1
Different approaches of data conceptualization process using ML.

No.	Author	Conceptual model
1	Hua et al. (2016)	McMc conceptual model
2	Arpan Kumar et al. (2018)	SEM model
3	Zhao and Mao (2017)	FBoWC model
4	Li, Sun et al. (2015)	Stickness function model
5	Liu and Yan (2016)	LCPL similarity model
6	Li et al. (2015)	SLCA / MI model

2. Revisiting theoretical contributions on ML and Data driven methodologies

During the semantic analysis and organization process of IR, the conceptualization webs a conceptual network for the query to move on some specific semantic paths (Liu and Yan (2016), Liu et al. (2015), Tran and Phan (2017), Kaur and Aggarwal (2018), Papadimitriou et al. (2017)). The descented clued modification of the Concept Network (CN) then drives the query. The process of identifying the concepts or related topics based on the statistical analysis of the query segments is called the Latent Semantic Analysis (LSA) and the same in more semantically relevant terms is known as Latent Semantic Indexing (LSI). The LSA/LSI uses Singular Value Decomposition (SVD) approach to decompose the query-CN matrix (Onifade et al. (2018), Howland and Park (2004)). An another approach of concept production, called Latent Dirichlet Allocation (LDA), the Dirichlet distribution is utilized to obtain more restrictive Gaussian assumptions. The Probabilistic Latent Semantic Analysis (PLSA) and the Latent Dirichlet Allocation (LDA) approaches are widely used for unsupervised topic analysis in data mining applications. The independent uncorrelated feature selection approach such as the Principal Component Analysis (PCA) is incorporated with the PLSA and LDA. The PCA performs the eigen decomposition of the covariance matrix of the features and hence reduces the dimensionality issues. The effective information reconstruction is possible with the hybrid use of the SVD and PCA. The FCM validated queries, and the derived intention clusters can be used for enhanced word embedding approaches. The skip-gram based word2vec is a prediction model used in the word embedding approach to predict the context word vectors from the given concept vectors (Li et al. (2015), Murthy S. (2017), Blei et al. (2003), Khan et al. (2017)).

Table 1 lists out some of the recent works in data conceptualization and semantic data analysis to design the enhanced search ontology.

There are three types of conceptual approaches are seen in keyword search using conceptual graphs. First one is concepts measured in terms of average edge weight over a compressed concept network for the query called conceptualization based on average edge weight (Yu et al. (2016), Huang et al. (2017), Chien and Chueh (2011), Song et al. (2015)). The conceptual edge weight is calculated based on various parameters such as semantic coherence, semantic relatedness, semantic similarity and using co-ranking frameworks. The occurrences of shared concepts make the query more semantically meaningful. The symmetric similarity approach such as cosine similarity with TF-IDF is widely used to represent the conceptual edge weight in language processing. The conceptual short text embedding models are useful to learn and predict the conceptual representation (Kusner et al. (2015), Li et al. (2019), Venugopal et al. (2009), Sijin and Champa (2020), Chiang et

al. (2015), Singh et al. (2022)) for the above approach. The FCM conceptual rings make these concepts more tightly packed and guaranteed to obtain a precise context of the search. The second one is conceptual semantic search diversification approaches. Here, the query terms and their semi features are considered for calculating the segment weight of the query. The context of the search is identified by the mutual evaluation of the relevance and novelty of the query. In XML, data trees, this approach proceeds with Smallest Lowest Common Ancestor (SLCA) / Extendet Lowest Common Ancestor (ELCA) based tree pruning (Li et al. (2015), Vieira et al. (2011)). Approximate search is incorporated with the second approach to avoid the empty results Liu and Yan (2016). In database oriented methods, the diversification is achieved through data annotation and the applications of fuzzy threshold algorithms Jain and Ipeirotis (2009). The third one is local and global context level conceptualization (LGCC) which is a linguistic and Natural Language Processing (NLP) based approach and acts as a supporting model for the first two approaches. It points to the importance of global and local validity of the search Li et al. (2012). Popular LGCC approaches are Named Entity Recognizer (NERs), POS taggers, and other Natural Language Processing (NLP) algorithms.

The crucial insights about the hypothetical data proliferates the data in hand. The data is growing and enriching with newer contents and needed artifact to discover the potential pattern of modern economic knowledge Zhang et al. (2009) or Information. The Mathematical model, Rough set model, and Fuzzy set models are the popular examples to learn from such complex data proliferation. The theories of Theory building not only restricted to the field of Information retrieval but also to various ML applications. The works of Chauhan and Palivela (2021) provide a deep learning approach to prevent fake news propagation. Their predictive ML model utilized the method of vectorization, and tokenization for conceptualizing the news segments. The word embedding approach is used to represent related words or intended data about a fact on social media. The hybrid feature selection model mentioned by Chaparala et al. (2023) is used to measure the strength status of building constructions. This kind of "prevention is better than cure" approach aligned on analysis of instances of the hypothetical core data, such as Animal behavior, water level, temperature, velocity of P and S waves to build up the theory of an Earth quake monitoring system.

3. Methodology

Usually the search diversification systems compute the top-k qualified search results by balancing the relevance and novelty of the derived query results. The relevancy guarantees the percentage of similarity between the given query and the derived query. The novelty is the percentage of newness between the given query and the generated query.

3.1. Problem statement

For the given query q (q_1, q_2,....q_n) and the given data network G (V, E), form a conceptual network G' (V', E') by the typed terms of q. Identify the best average edge weights Avg(EW) = $\frac{\sum_{e \in E} w(e)}{n(E)}$ between (u, v), where $u, v \in G'$ and u! =v. Perform the feature analysis of the obtained features by calculating their mutual information score. The appropriate contextual meaning of the query is defined by the SLCA region encoding of the diversified anchors on the XML tree, XDT.

3.2. Conceptualization with semantic search diversification

The baseline anchor pruning algorithm is proceeding with the type checked query segments. Type checking is the process of finding the best collocation of a query segment on the basis of Singleton Score (SS). The SS is the lexical representation of query terms. It is obtained by mapping the type of query terms with corresponding POS tags. The edge weight of the query is calculated by the fuzzy logical combination of Affinity Score with Singleton scores as given in (1).

Table 2
Feature Extraction of the user query "Binary Search".

Keyword	Features
Binary	search, optimum search, tree
Binary	optimum search, operation, Data structure
Binary	multidimensional, binary tree applications
Search	nets, search, Huffman, rational
Search	range, multi-way, parse, balanced
Search	B+ tree, quad tree, problem, evaluation

$$W(a,b) = SS(a).AS(a,b).SS(b) \tag{1}$$

The Affinity Score is calculated as the cosine similarity between the concept vectors of the query segments as given in (2).

$$AS = cosine\left(a\left(\overline{CV}\right), b\left(\overline{CV}\right)\right) \tag{2}$$

The Table 2 lists out some of the important features obtained for the query segment "Binary Search" after type detection or type checking. These results, and their respective instances are used to formulate the intention cluster to obtain the derived results. The intention cluster guarantees the optimum results, reduces the number of results for further computations and avoids overwhelming situations. The mutual information values of the obtained feature terms with the given keywords are used to determine the contextual measure and relevancy level of the new candidate queries. The MI (x, y) is used to represent the co-occurrence and semantic relation of the given keyword segments and query features and is given in (3) and the values are listed in Table 3, where x and y are two discrete random variables.

$$MI(x,y) = P(x,y) + log\frac{P(x,y)}{P(x).P(y)} \tag{3}$$

The semantic interpretation of the query q is represented by a variable weight(q_{new}) based on the relevance and novelty of the generated query.

$$weight(q_{new}) \propto \frac{\bigcap qft_i \in q_{new} R(qft_i, DT)}{R(DT)}$$

Where $\bigcap \frac{R(qft_i, DT)}{R(DT)}$ is the SLCA nodes of the query with it's semi-features and qft_i is the generated query formed by the association of any of the query keywords with it's corresponding semi-features.

4. Feature extraction and mutual information

The Fig. 1 shows a typical Conceptual search diversification system based on the MI. The word embedding on the Table 3 results is conducted by referring the intention cluster to the XML tree.

The proposed Anchor based Evaluation of Semantic Pattern algorithm lists out the required query matches and it's partial matches as mentioned in section 4.2. Then the algorithm calculates the flat probability of the query and the generated query based on the size of the node list obtained.

A feature vector is used to hold the relevant features of the query keywords and hence formed a reduced set of associated words. These values reflect the mutual dependency between the feature vector and the query terms in their various contexts called search diversification.

4.1. Factors affecting the search diversification process

The feature extraction process generates more keywords which are having equal tendency to divert the query if their path length such as Longest Common Path Length (LCPL) is attributed to some common concepts. The newly generated queries can be modeled by an aggregated score function as given in (4).

Table 3
Mutual information values of various contextual terms with respect to the query terms.

Keyword	Features	MI value	Keyword	Features	MI value
Binary	search	9.28	Tree	Huffman	7.97
Binary	testing	7.97	Tree	search	7.64
Binary	tree	7.28	Tree	Y-Tree	7.97
Binary	data structure	8.7	Tree	tree net	5.64
Binary	operation	7.7	-	-	-

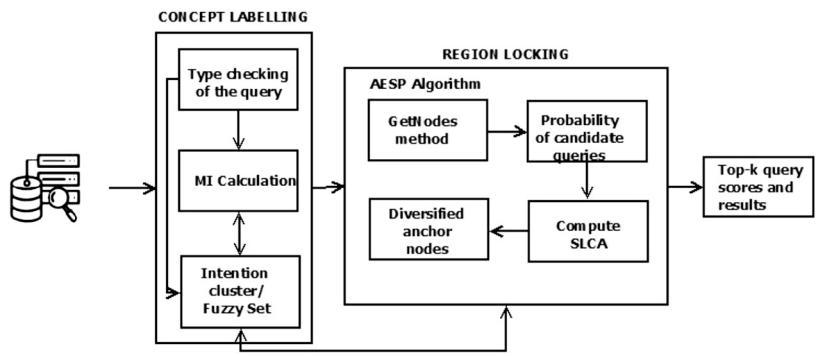

Fig. 1. Conceptual search diversification process.

$$weight(q_{new}) = P(\frac{q_{new}}{q}, XDT) * NDIF(q_{new}, Q, XDT) \tag{4}$$

Where $P(\frac{q_{new}}{q}, XDT)$ is the probability of generating a new semantically related query for the given query q. The second term, NDIF (q_{new}, Q, XDT) is the measure of difference between the newly generated queries from the previous queries.

4.1.1. Evaluating the semantic similarity and novelty

Words with the same meaning have similar representation in a word corpus is the working principle of Word embedding methods (Li et al. (2017), Onan (2019)). In order to achieve such a representation, the first part of the (4) is modeled as in (5).

$$P(\frac{q_{new}}{q}, XDT) = \frac{P(\frac{q}{q_{new}}, XDT) * P(\frac{q_{new}}{XDT})}{P(\frac{q}{XDT})} \tag{5}$$

Where $P(\frac{q}{q_{new}}, XDT)$ is the probability of getting query q as a coherent term for generated query q_{new} in the given XML data tree XDT. $P(\frac{q_{new}}{XDT})$ is the query generation probability of the system. $P(\frac{q}{XDT})$ is the probability of the system to search for the given query. It is possible to rewrite $P(\frac{q}{q_{new}}, XDT)$ as in (6).

$$P(\frac{q}{q_{new}}, XDT) = \pi_{q_i \in q, ft_{ij} \in q_{new}} P(\frac{q_i}{ft_{ij}}, XDT) \tag{6}$$

Where $P(\frac{q_i}{ft_{ij}}, XDT)$ is defined as the probability of query term correlated with the given feature and is represented as qft_i for example q = 'Binary' and feature term is 'search'. It is interpreted as follows and given in (7). The query generation probability $P(\frac{q_{new}}{XDT})$ can be calculated and given in (8). The $\bigcap qft_i \in q_{new} R(qft_i, XDT)$ is the SLCA values of query term with it's features. From the above knowledge the (5) can be rewritten as in (9) where $\frac{1}{P(\frac{q}{XDT})}$ has a value between $(0, 1]$ and is denoted as β.

Novelty is the process of incrementally refine the diversified results into more specific one and is given in (10). The process of evaluating the novelty for a newly generated query candidate q_{new} depends on the evaluation of the other previously generated query candidates set Q.

The $R(q_{new}, XDT)$ is the SLCA values generated by q_{new}. $\bigcap q_{\cdot} \in QR(q', XDT)$ is the SLCA results generated by queries in Q. $nx \leq ny$ gives vertex nx is a duplicate or an ancestor of vertex ny. The denominator is the union of SLCA results generated by the query in Q and q_{new}. In terms of relevance and novelty (4) can be rewritten as shown in (11).

$$P(\frac{q_i}{ft_{ij}}, XDT) = \frac{P(\frac{ft_{ij}}{q_i}, XDT) * P(\frac{q_i}{XDT})}{P(\frac{ft_{ij}}{XDT})}$$

$$P(\frac{q_i}{ft_{ij}}, XDT) = \frac{R(\frac{q_i, ft_{ij}, XDT}{R(XDT)})}{R(\frac{ft_{ij}, q_i, XDT}{R(XDT)})}$$

$$P(\frac{q_i}{ft_{ij}}, XDT) = \frac{R(qft_i, XDT)}{R(ft_{ij}, XDT)} \tag{7}$$

$$P(\frac{q_{new}}{XDT}) = \frac{R(q_{new}, XDT)}{R(XDT)} = \frac{\bigcap qft_i \in q_{new} R(qft_i, XDT)}{R(XDT)} \tag{8}$$

$$P(\frac{q_{new}}{q}, XDT) = \beta * \frac{R(qft_i, XDT)}{R(ft_{ij})} * \frac{\bigcap qft_i \in q_{new} R(qft_i, XDT)}{R(XDT)} \tag{9}$$

$$Novelty(q_{new}, Q, XDT) = \frac{\{nx | nx \in R(q_{new}, XDT) \bigwedge \nexists ny \in \{\bigcap q' \in QR(q', XDT)\} \bigwedge nx \leq ny\}}{R(q_{new}, XDT) \cup \{\bigcap q' \in QR(q', XDT)\}} \tag{10}$$

$$weight(q_{new}) = \beta * \frac{R(qft_i, XDT)}{R(ft_{ij})} * \frac{\bigcap qft_i \in q_{new} R(qft_i, XDT)}{R(XDT)} * \frac{\{nx | nx \in R(q_{new}, XTD) \bigwedge \nexists ny \in \{\bigcap q_{\cdot} \in QR(q', XDT)\} \bigwedge nx \leq ny\}}{R(q_{new}, XDT) \cup \{\bigcap q_{\cdot} \in QR(q', XDT)\}} \tag{11}$$

$$weight(q_{new}) = \frac{\beta}{R(XDT)} * \frac{R(qft_i, XDT)}{R(ft_{ij})} * \frac{\bigcap qft_i \in q_{new} R(qft_i, XDT)}{} * \frac{\{nx | nx \in R(q_{new}, XTD) \bigwedge \nexists ny \in \{\bigcap q_{\cdot} \in QR(q', XDT)\} \bigwedge nx \leq ny\}}{R(q_{new}, XDT) \cup \{\bigcap q_{\cdot} \in QR(q', XDT)\}}$$

$$weight(q_{new}) = \frac{R(qft_i, XDT)}{R(ft_{ij})} * \frac{\bigcap qft_i \in q_{new} R(qft_i, XDT)}{} * \frac{\{nx | nx \in R(q_{new}, XTD) \bigwedge \nexists ny \in \{\bigcap q_{\cdot} \in QR(q', XDT)\} \bigwedge nx \leq ny\}}{R(q_{new}, XDT) \cup \{\bigcap q_{\cdot} \in QR(q', XDT)\}}$$

4.2. Semantic search diversification algorithm

The semantic search diversification is achieved through AESP. The algorithm assigns each processor a region for SLCA computation. Based on the obtained SLCA node called anchor node, four regions are considered called $anchor_{pre}$, $anchor_{next}$, $anchor_{anc}$, and $anchor_{des}$. The $anchor_{anc}$ is not included in the further SLCA computation due to the exclusive property of the SLCA. The new queries q_{new} and their corresponding semi features are maintained in a vector V. When the new query is processed, the information about shared segments and the necessary communication details are maintained among processors. If the new query candidate contains shared parts (features) called ψ in V, for each shared part, a processing status is set once it is processed. In another query search the obtained information can be passed to the required processor. The anchor pruning process avoids a lot of nodes without computation, and parallelism reduces the computation time.

It is seen from the algorithm that the FeatureAnalysis() function formulates a 2D feature table consists of the query terms and their fuzzy semantic parts. The features of all these items are identified and stored in the feature table FT_{ij}. The GenerateNewQuery function is invoked to obtain new queries q_{new}. The SLCA nodes are identified for each query term with its corresponding semi-features (Binary Search, Binary Tree, etc.).

P_qft measures the probability of partial query instances over the XDT and P_qnew is to represent the probability of the new candidate query with the presence of the given query over the given XDT. The SLCA nodes are identified for the threshold semi features (Binary tree, XML tree etc.). The Anchor nodes $node_{anchor}$ are identified for each SLCA listed group.

The word embedding of the obtained SLCA nodes with the intention cluster of the given query (e.g. Binary Search) is used to avoid the computation of the unwanted and unimportant SLCA nodes. A tree partition function called Partition($qins_{i,j}$,$node_{anchor}$) is used to perform partition on various directions called $qins_{pre}$, $qins_{dec}$, $qins_{next}$.

The SLCACalc($qins_{pre}$,$node_{anchor}$) function is used to calculate SLCA nodes of the previous region of the anchor. The SLCACalc($qins_{des}$,$node_{anchor}$) is used to calculate SLCA nodes for descendants. If the result set ϕ" is empty, then the remove method is used to remove the $node_{anchor}$ node in order to maintain the exclusive property of the SLCA. In an another query search of the same instance, the obtained information can be passed to the required processor as a part of parallelization of the AESP algorithm. Because of space limitation, the details of the parallelization are not included in this paper.

Algorithm 1: Anchor based Evaluation of Semantic Pattern algorithm-AESP.

Data: A search query q with n keywords, it's Fuzzy set FS and an XML data tree XDT.
Result: Top-k query intentions Q_{top} and overall result set ϕ.
initialization;
1. FT_{mxn} = FeatureAnalysis(q, FS, XDT);
2. while q_{new} = GenerateNewQuery(FT_{mxn}) $\neq$ null do
3. ϕ = null and P_qft{i, j} = 1;
4. $qins_{i,j}$ = SLCANodeList($qft_{i,j}$, XTD) for $qft_{i,j} \in q_{new} \wedge 1 \le i \le m \wedge 1 \le j \le n$;
5. P_qft = π $ft_{ij} \in qft_{ij} \in q_{new} \frac{qins}{SLCANodeSize(qft_{i,j}, XTD)}$;
6. ϕ = SLCACalc($q_{i=1 \text{ to } n}$)*$|FS|$ and P_qnew = P_qft{i,j}*ϕ
7. If ϕ is not empty then
8. For all $node_{anchor} \in \phi$ do
9. Perform Partition($qins_{i,j}$,$node_{anchor}$) to $qins_{pre}$, $qins_{dec}$, $qins_{next}$
10. if $\forall$ $qins_{pre}$! =null then
11. ϕ' = SLCACalc($qins_{pre}$,$node_{anchor}$);
12. if $\forall$ $qins_{des}$! =null then
13. ϕ" = SLCACalc($qins_{des}$,$node_{anchor}$);
14. $\phi = \phi + \phi' + \phi''$;
15. if ϕ" ! = null
16. ϕ.remove($node_{anchor}$);
17. if $\exists$ $qins_{next}$ = null then
18. Break the for loop;
19. qins = $qins_{next}$ for $1 \le i \le m \wedge 1 \le j \le n$; else
20. SLCACalc($qins_{ij}$);
21. weight(q_{new}) = $P_q_new * \phi * \frac{\phi}{\phi + \phi'}$;
22. if $|Q| \le$ k then
23. put q_{new}:weight(q_{new}) into Q;
24. put q_{new}:ϕ into ϕ;
25. else if weight(q_{new})>weight($q_{new'}$)
26. replace $q_{new.}$:weight($q_{new.}$) with q_{new}:weight(q_{new});
27. ϕ.remove($q_{new'}$);
28. Return Q and ϕ;

5. Performance evaluation

5.1. Effectiveness of type detection

The amount of impact lexical feature θ has an important role in determining the precision of type detection and it ranges from [0-1]. Both Chain Model (CM) and Pairwise Model (PM) show comparatively high precision when $\theta = 0$ Hua et al. (2016). It is able to note a steep rise in precision when θ is in the range of [0-0.1] where both semantic and lexical types are considered. When θ value further increases precision deteriorates because lexical features impose linguistic conceptualization over a vocabulary. Fig. 2 shows the results for the CTTQ in which both tweet and query terms show a high level of precision. The CTTQ conceptual phase works on concept vectors of terms over CN and formed a semantic world with concepts and their instances on a pairwise model to get a new weight which is on the line of data conceptualization rules.

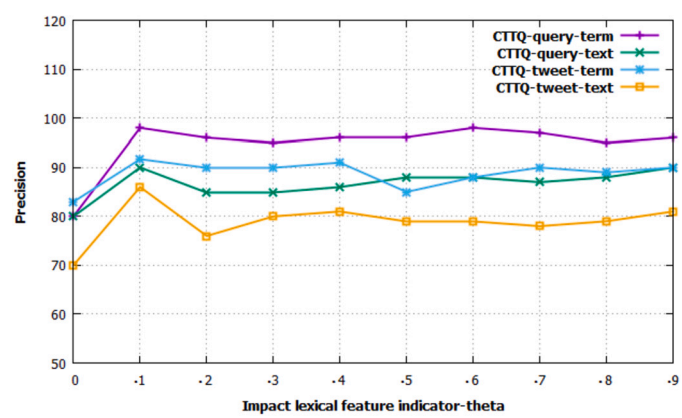

Fig. 2. Precision of type detection when θ varies.

5.2. Effectiveness of the search diversification algorithms based on query suggestions

The CTTQ's intention cluster approach provides all the necessary instances at a time, and the SLCA locking is achieved at once. The partial matching process of the top-k query intentions provides more diversification of search to the desired users similar to the facets. The n-DCG measures reveals that, the average response time for calculating the SLCA results for a query from the given query suggestions using Anchor based Parallel Sharing (ASPE) algorithm with XRank and proposed AESP is shown in Fig. 3a. The experimental results show that AESP shows considerably less response time i.e. 20 percent time of baseline methods with five suggestions, 33 percent time of baseline methods with ten suggestions, 38 percent time of baseline methods with fifteen suggestions, 34 percent time of ASPE with twenty suggestions. When the query suggestions are more, there should be more SLCA computations with these suggestions and response time increases accordingly.

In the case of short and vague queries, the AESP has to process comparatively less features and the SLCA nodes wanted to process are further reduced. Which in turn shows a considerable reduction in response time with the least number of query suggestions in the case of

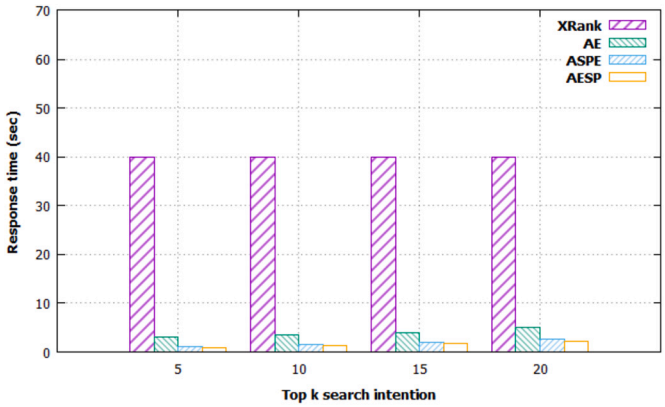

Fig. 3a. Average response time.

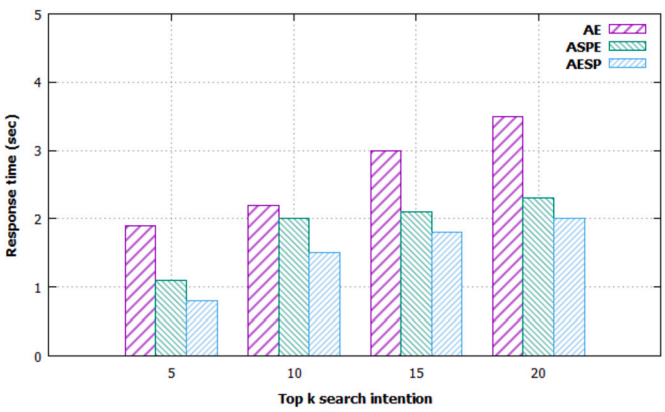

Fig. 3b. Average response time (Vague).

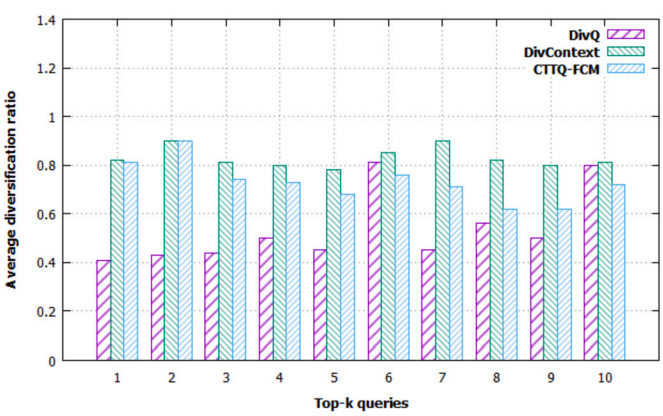

Fig. 4. Differentiation of diversified query result.

the proposed model as shown in Fig. 3b. In the Maximal Clique by Monte Carlo (MCMC) and the Maximal Clique with Cycle Count (MCCC) Sijin and Champa (2021) evaluated query over the FCM Zhao and Mao (2017) trained corpus, the response time is still more reduced.

5.3. Test differentiation of diversified query results

The experimental results show that the CTTQ diversification model provides significantly better differentiated results along with the existing DivQ Demidova et al. (2010) and DivContext Li et al. (2015) approaches. In the case of an idle diversification system design, the structural difference between the generated structured queries has to reflect the diversity of the results. It has to consider both the structure and the word context of the newly generated queries. In such an approach if the search focuses on any particular levels of the XDT (e.g. node Article in DBLP), the specific keyword and its word context in the correlated graph will be locked and reveals the results as in the case of SLCA region locking.

The CTTQ identifies the specific keyword and applies SLCA locking of the keyword with partial matches to achieve diversification, but its conceptual graph is instance based and the SLCA locking covers the intention cluster. Thus, the CTTQ is able to produce diversified results within the scope of the intention cluster and avoids flooding and structural dissimilarity without any structural modifications and is shown in Fig. 4.

6. Discussion

6.1. Contributions to literature

The confident query segment Hua et al. (2016), Li, Sun et al. (2015) evolved through text segmentation, type detection and concept labeling derive a contextually measured query peace of user interest. The proposed MCCC algorithm considers the conceptual rings and shared concepts to produce a more valuable query peace on any cohesive Concept network such as SEM and SEO. Missing the potential customers on Search Market (SM) will bring tragedy to the ongoing industry is the foresight of Arpan Kumar et al. (2018). If SEM and SEO are engaging their customers by approximating on keyword search (Liu and Yan (2016), Jain and Ipeirotis (2009)) and adjusting on advertising budgets, they can concern user generated content and can avoid the “absence of match box on a crackers festival” situation.

Efficient query based region encoding is possible with the distribution of SLCA anchor nodes (Li et al. (2015), Vieira et al. (2011)) on XML Data tree. The CTTQ refers the intention cluster during the region encoding and avoids the wasted computations of the unwanted semi-features. There are many vendors and they have products of many industries to market on. The digital marketing firms can adopt any business strategies to advertise it by using various Data mining services. The data conceptualization and its surrogates Zhao and Mao (2017), Papadimitriou et al. (2017), Ruiz et al. (2014) are one of the best approaches to measure the user concerns and its numeric representation.

6.2. Implications for practice

The CTTQ is a Datamining service to measure the user intentions using the data conceptual approach of Data driven paradigm. It is designed to use with the existing Search services like SM, SEM and SEO generally and other web services like SEOClerks, Fiverr, Upwork, Umso, Gigbucks, in which the user concern is a direct prime factor for running the business and the industry. The conceptual rings and shared concepts provide a modified query of user concerns and intentions. This wanted concepts collectively forms an intention cluster to guide the query on any new regions of the Concept network. The precise numerical measures of the query are useful in many data mining and allied services like marketing, supply chain management and social network analysis. It offers a better information management prerequisite to researchers as well as corporate vendors in data clustering, classification, and other data analysis and management processes.

Since the existing SLCA based region encoding or any such approaches can able to lock the user needs, it has limitations in many aspects like the number of keyword matches, its operations, and unwanted data pruning. If the intention cluster is incorporated during the SLCA, ELCA or LCPL calculations, only the intended concepts are used for SLCA node calculation operations, so a word embedding approach is more than enough to lock only the wanted SLCA nodes. In XPath, XQuery based IR system development process, the proposed fuzzy approach can effectively use.

7. Conclusion

Identifying the contextual meaning of a search query from its candidate terms is the turning point of a search diversification system development process. The proposed Semantic Search Diversification method identifies the contexts of the input query by balancing the relevance and novelty of the queries. The word embedding of the query on the XML data tree is used to identify the desired SLCA nodes. A feature vector is used to hold the relevant features of the query keywords and hence formed a reduced set of associated words on the concept network. The anchor nodes are used to localize other diversified results of the query. The Anchor based semantic pattern matching algorithm proceeds to the left, descendant and next levels of branches of the tree to list out the new top-k SLCA nodes for each assigned region. The nDCG measures of the proposed work show more than 75 percent of chance to approach the ground truth of the ideal diversified query suggestions. High usefulness is achieved even for vague queries with the least number of relevant and novel query suggestions. Apart from nDCG the impact lexical feature indicator is also used to bias the relevancy of the search results.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jjimei.2023.100196.

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